Input system:
Dependent variables:x_1
Independent variables:t_1, t_2, t_3
Equations:

$$\frac{\partial x_1}{\partial t_1} = x_1 \sin(x_1)$$

$$\frac{\partial x_1}{\partial t_2} = x_1$$

$$\frac{\partial x_1}{\partial t_3} = 5$$

==========

Result:
Dependent variables:s_1, x_1, s_2
Independent variables:t_1, t_2, t_3
Additional variables:

$$s_1 = \sin(x_1)$$

$$s_2 = \cos(x_1)$$

Equations:

$$\frac{\partial x_1}{\partial t_1} = x_1 s_1$$

$$\frac{\partial x_1}{\partial t_2} = x_1$$

$$\frac{\partial x_1}{\partial t_3} = 5$$

$$\frac{\partial s_1}{\partial t_1} = s_2 x_1 s_1$$

$$\frac{\partial s_1}{\partial t_2} = s_2 x_1$$

$$\frac{\partial s_1}{\partial t_3} = 5 s_2$$

$$\frac{\partial s_2}{\partial t_1} = (-1) s_1^2 x_1$$

$$\frac{\partial s_2}{\partial t_2} = (-1) s_1 x_1$$

$$\frac{\partial s_2}{\partial t_3} = (-5) s_1$$